

Evaluation Techniques for Modern Military Camouflage

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Abstract—The integration of AI in military operations has enhanced the ability to detect camouflaged objects, as evidenced by the increased use of drones in conflicts such as those in the Middle East and the Ukraine-Russia war. This shift towards AI-powered detection systems necessitates a re-evaluation of camouflage evaluation methods and metrics.

This systematic literature review examines the evolution of camouflage techniques from naturalistic patterns to advanced designs taking into account AI and drone technology. It explores both traditional and modern evaluation methods, focusing on their applicability in military contexts. The study employs a mixed-methods approach, combining qualitative and quantitative analysis to provide a comprehensive evaluation of camouflage effectiveness. It also incorporates context-specific and generalisable metrics to ensure thorough evaluation across different environments and scenarios.

The findings highlight the importance of developing robust evaluation techniques that can address the challenges posed by both human and AI detection systems. This review underscores the need for continuous improvement in camouflage design and evaluation to maintain the effectiveness of military strategies in modern warfare.

Keywords— Defence Technology, Camouflage pattern evaluation, military camouflage, AI.

I. INTRODUCTION

Camouflage plays an important role in military strategy, enabling military assets to blend into the environment and avoid detection. Effective camouflage design is a crucial element attributing to the success of military operations and the overall safety of personnel [1]. With the increasing capabilities of computational algorithms and artificial intelligence (AI) models, the necessity to develop effective camouflage designs that can deceive both humans and AI systems has significantly grown. Military operations are increasingly incorporating AI, which significantly boosts the detection capabilities for camouflaged objects [2]. As per example, the number of drone attacks launched in the ongoing conflict in the Middle East underscores the growing reliance on AI-powered systems in modern warfare. Similarly, the Ukraine-Russia conflict has highlighted the increased use of drones compared to the start of the most recent conflict in 2022, with continuous drone attacks and countermeasures

becoming a common aspect of the modern battlefield. The transition to ground drone operations underscores the importance of revisiting camouflage evaluation methods and metrics, as today's observers extend beyond human capabilities. Being able to adequately evaluate camouflage designs in the face of both human and AI detection is therefore essential.

The study of camouflage effectiveness encompasses a wide range of disciplines, including biology, psychology, engineering, and computer science. Historically, camouflage techniques have evolved from simple naturalistic patterns to sophisticated designs that incorporate advanced materials and technologies [3]. Recent advancements in AI and drone technology have further enhanced the ability to design and evaluate effective camouflage. AI-powered systems equipped with high-resolution cameras and sensors are increasingly capable of detecting camouflaged objects, posing new challenges and opportunities for camouflage design and evaluation [4], [5]. This systematic literature review builds on existing knowledge by examining both traditional and modern evaluation methods and metrics, with a focus on their applicability in military contexts. Various evaluation methods are explored that can be applied across different spectra, with a particular focus on camouflage evaluation in the visible spectrum.

The following sections explore various aspects of camouflage evaluation. Section II delves into the historical evolution and recent technological advancements, particularly in AI and drone capabilities. Section III outlines the research methodology. Section IV discusses the evaluation methods, both qualitative and quantitative. Section V categorises the evaluation metrics into generalisable and context-specific metrics. Finally, Section VI summarises the key findings.

II. REVIEWED RESEARCH CAMOUFLAGE EVALUATION

The development of military camouflage mechanisms can be seen as a part of the ongoing developments and challenges existing in warfare, and due to the increased sophistication of detection technologies. In its earliest forms, camouflage relied on simple methods such as using natural colours and materials that would help soldiers and their equipment to blend into various surroundings. Nevertheless, given recent

and current technological advancements, new intelligent and adaptive methods are being developed like the ones that break up recognisable shapes and multispectral designs capable of masking visibility across visual, infrared, and radar spectrums [3]. An important set of solutions is developed using various AI techniques, given their capabilities to form and recognise patterns, engage in deception, and adapt in real-time to various environments. Such solutions allow countering modern threats such as unmanned aerial systems and high-resolution imaging tools, enhancing stealth and survivability in increasingly complex operational landscapes [6].

Military camouflage is based on diverse patterns and technologies designed to enhance concealment and adapt to varying environments. Terrain-specific patterns, such as woodland, jungle, and desert designs, optimise concealment by blending with local colours, like green and brown tones for forests and tan hues for deserts. Universal camouflage patterns that were introduced to standardise camouflage, faced criticism for their inefficacy in diverse environments like Afghanistan due to their bright, non-earth-tone colours. Transitional patterns represent a compromise, performing adequately across multiple terrains when environment-specific options are impractical. In recent advancements that include MultiCam, have demonstrated superior performance in blending and detection resistance through rigorous field and photo-simulation testing, making them ideal for modern operational conditions [7]. In NATO, military camouflage is not only a tool for concealment, but a symbol of operational cohesion, modernisation, and strategic adaptability based on interoperability across diverse environments while addressing modern challenges, such as visual, infrared, and radar-based detection [8]. Furthermore, in Ukraine, military camouflage plays a double role as it is important in both defensive operations and as it is a symbol of collective resistance against external aggression. Techniques such as camouflage netting are used to conceal personnel, equipment, and cultural landmarks from advanced reconnaissance technologies, including aerial drones and thermal imaging. These nets, are made using discarded textiles, including military uniforms and civilian materials, to create effective concealment solutions [9].

To assure the effectiveness of military camouflage patterns and solutions in reducing detectability and enhancing the survivability of military personnel and equipment in operational environments, the evaluation phase needs to be conducted in a structured, systematic, and clear manner. This implies assessing how well camouflage patterns and materials blend with their surroundings, helps identify weaknesses in camouflage designs, informs plans of improvements, and ensures robustness against modern detection technologies. The evaluation process implies subjective, objective, and computational methods to assess its effectiveness in minimising detectability. Subjective methods include observer experiments, where experts assess camouflage in controlled scenarios or real-world settings, using metrics such as detection probability and search times. Objective methods employ physical measurements, like spectral reflectance and emissivity, to quantify how well camouflage materials blend with their surroundings across visible and infrared spectra. Computational methods use image processing and predictive models like saliency maps and target-background similarity metrics to simulate and evaluate camouflage performance [5].

[10] uses the CSI (Camouflage Similarity Index) metric designed to quantify the blending effectiveness of camouflage designs with their backgrounds. Images of camouflage and backgrounds were processed into uniform pixel dimensions, and their colour values were analysed in the CIELAB colour space using MATLAB. Further, post-hoc statistical tests were conducted to determine significant differences between designs. The CSI effectively identified blending performance variations, enabling categorisation of designs based on effectiveness. In a similar study, [11] proposes the MF-CFI (Multi-Feature Camouflage Fused Index) to assess the similarity between camouflage patterns and their backgrounds. The study involves eye movement experiments where participants observe digital and large-spot camouflage patterns in simulated environments. Eye tracking data, including hit rates, fixation duration, and saccade amplitudes, are used to validate the MF-CFI's effectiveness. The findings indicate a high correlation between the MF-CFI and visual detection performance, making it superior to traditional metrics like CSI.

Based on deep learning techniques, [12] conducts the evaluation phase in two steps: 1) a GAN-based model to detect anomalies between target and background features to distinguish high-performance camouflage and 2) a MobileNetV2 model to evaluate target feature similarity when classifying camouflage levels. Furthermore, [13] builds a camouflaged object detection solution based on the SINet-V2 algorithm which is assessed considering S-measure, E-measure, weighted F-measure, and Mean Absolute Error (MAE) metrics. This method demonstrates the potential of tailored datasets and advanced deep learning in improving military camouflage object detection performance.

Nevertheless, to enhance both operational effectiveness and decision-making transparency, AI-based military camouflage solutions need to be developed, deployed, and used in an explainable and trustworthy manner. Explainability enables military experts to understand how AI systems assess camouflage patterns, ensuring decisions align with human perception and tactical requirements. System trustworthiness, on the other hand, reduces risks associated with biases or errors in evaluating camouflage performance across diverse environments, enhancing reliability in real-world deployments.

This approach supports and enhances confidence, accountability, and collaboration between human experts and AI systems, further improving the quality and effectiveness of military camouflage solutions [14].

In an adversarial approach, [15] evaluates military camouflage effectiveness through adversarial patch attacks to deceive automated aerial detection systems using the YOLOv2 object detection model. Evaluation metrics, including precision-recall curves and average precision, were used to compare the effectiveness of adversarial patches against baseline noise patches. The results demonstrated that adversarial patches significantly reduced detection rates and outperformed traditional noise patches. This approach aims to exploit vulnerabilities in object detection models to evade aerial surveillance systems, being an AI deception mechanism, thus an AI security issue. Furthermore, it emphasises the need for enhanced defence mechanisms that include adversarial training and robust feature engineering, to mitigate such attacks [16]. This not only exemplifies offensive

AI tactics, but also serves as a reason for developing more resilient AI systems in military applications.

III. RESEARCH METHODOLOGY

The objective of this research is to provide an overview of existing evaluation methods and metrics used when designing military camouflage patterns in order to assure the operational and technological effectiveness of military camouflage solutions that military systems and personnel could use in various military environments. On this behalf, two research questions are formulated, as follows:

- Research Question 1: Which evaluation methods are used for assessing the effectiveness of military camouflage design patterns?
- Research Question 2: Which evaluation metrics are used for assessing the effectiveness of military camouflage design patterns?

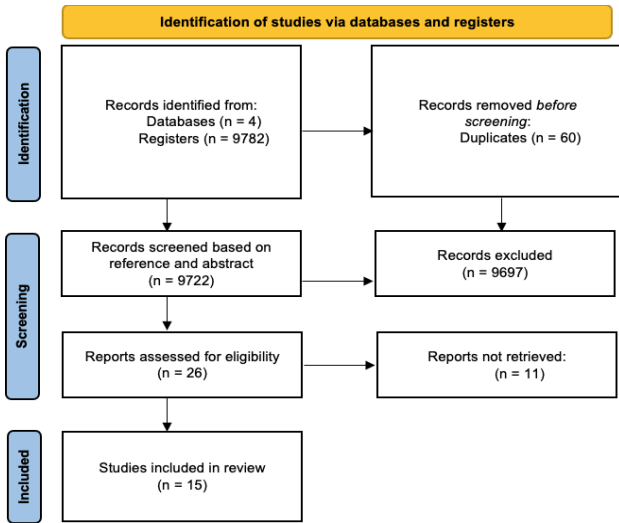


Fig. 1. Research process.

To address these questions and achieve the goal of this research, given the existing knowledge gap identified in the previous section of this study, a systematic literature review is conducted being guided by the guidelines and requirements established by the PRISMA methodology [17]. Accordingly, relevant research studies were identified and selected based on the criteria established by the research team. These studies were thoroughly analysed to address the research questions above presented, with the findings being synthesised, analysed, and presented in detail in this present study. To this end, search strings were constructed in order to query the IEEE Explore, ACM Digital Library, Wiley, and Taylor & Francis databases. The studies include only conference and journal articles published in English between 01.01.2019 and 30.11.2024.

This search resulted in 9782 studies that contained 60 duplicates. Further, based on reading the abstracts, titles, and keywords, and applying the inclusion and exclusion criteria considered which implies to refer to studies that are dedicated to camouflage patterns in the military domain, 9722 studies were integrated initially. From an in-depth analysis, 9697 studies were removed. At the end, 15 studies were considered to be included in the final step representing the pool of studies considered in this research. The overview of this process is depicted in Figure 1.

Furthermore, the research studies considered for this literature review are detailed in Table I.

IV. EVALUATION METHODS

To structure the evaluation methods analysed, two criteria are considered: nature and background. The nature criteria clusters the evaluation methods as being qualitative or quantitative where (i) quantitative methods use objective measurements and computational modelling based on metrics such as detection accuracy, loss rates, or spectral properties, and (ii) qualitative methods focus on human subjective evaluation and human perception considering user studies, visual assessments, and feedback from human participants to assess the usability, comfort, and effectiveness in real-world scenarios. The background criteria clusters the evaluation methods as being context-specific or generalisable where (i) context-specific methods are tailored to particular applications, e.g., in specific terrains, operational environments, or lighting conditions, often integrating domain-specific challenges like adversarial deception or terrain testing, and (ii) generalisable methods consider statistical analysis or loss function optimisation being adaptable to other fields. Each of these methods are further discussed taking into consideration that various studies addressed in this research consider a robust evaluation approach that integrates both these methods for balancing the quantitative precision with qualitative insights.

A. Quantitative methods

[18] uses a blend of methods such as recognition accuracy, lighting impact, vegetation coverage, and CNNs (Convolutional Neural Networks) prediction accuracy, loss rates, and correlation coefficients and simulations for urban and rural environments. Similarly, [29] employs loss function optimisation using MSCN (Mean Subtracted Contrast Normalized) coefficients and L-BFGS (Lighted-memory Broyden-Fletcher-Goldfarb-Shanno) algorithms, and statistical techniques for colour mapping. At the same time, [25] uses kinematic equations and hardware testing for energy efficiency and motor angles, supported by environmental testing for navigation and solar energy reliance. In a different approach, [20] applies eye movement metrics such as saccades and durations, texture comparisons, and 3D mapping evaluating texture patterns and clustering for dynamic colour extraction. [21] applies eye-tracking metrics like detection time and gaze duration together with statistical analysis to assess the challenges in camouflage detection. Moreover, in [23], the authors evaluate the reaction time using GLMMs (Generalized Linear Mixed Model) for statistical analysis of GAN (Generative Adversarial Networks) iterations and success assessment. In an adversarial setting, [22], includes content and style loss monitoring, colour space analysis and statistical testing to validate camouflage patterns. Furthermore, [30] proposes a quantitative evaluation methods based on the MSE (Mean Squared Error), Universal Image Quality Index (UIQI), objectiveness loss (LOBJ), and total variation loss (LTV) metrics. Moreover, [19] assesses the material durability and visibility reduction through computational modelling and detectability metrics, and [3] evaluates deception effectiveness with metrics like decoy success rates, resource misallocation, and imitation performance, measured in terms of agility, stealth, and functional equivalence.

B. Qualitative methods

[19] provides an overview of various methods that consider visual inspection of headgear and face paint, along with usability assessments which focus on user comfort and environmental adaptability, while real-world field tests offer practical insights. [22] focuses on visual assessments of camouflaged human-shaped targets, evaluating the concealment effectiveness of content, style, and hybrid patterns under specific conditions. At the same time, [27] uses visual inspections to assess the quality of camouflage blending in various settings, [30] uses visual inspections to evaluate the natural appearance and imperceptibility of adversarial patches, ensuring they blend effectively with backgrounds, and [31] employs visual inspections to confirm their compatibility with environmental surroundings. Moreover, [3] uses observational methods, including case studies like Operation Bertram, to evaluate the effectiveness of camouflage and decoys in mimicking biological traits and deceiving adversaries. With user studies, in [20], [23], feedback is provided on the difficulty of detecting camouflaged targets, offering insights into real-world applicability. Additionally, [24] incorporates the time required to locate the targets into the detection process. Moreover, [28] considers the detection of varying camouflage patterns and provides insights into pattern blending through comparisons with established designs like DPM (Disruptive Pattern Material). While combining human-centric evaluations with technical assessments, [21] uses qualitative methods like visual analysis of areas of interest (AOI) and participant studies with eye-tracking glasses to explore subjective interpretations of camouflage patterns in controlled settings. [25] combines functional testing with behaviour assessment to provide qualitative insights into operational performance. Meanwhile, [26] evaluates how well a robot's camouflage blends with its environment, relying on subjective human judgement to evaluate its effectiveness. Moreover, [29] evaluates the effectiveness of integrating camouflage patterns with environments, simulating human judgement of texture and colour blending.

C. Context-specific methods

Methods such as UAV-based recognition accuracy, lighting impact, and vegetation coverage [18] as well as environmental adaptation testing and usability assessments [19], are closely tied to evaluating camouflage performance in specific terrains. At the same time, eye movement metrics, user studies, and 3D mapping [20] focus on visual aspects and adaptability in various military environments. Techniques such as 3D camouflage mapping and AOI visual analysis [21] extend their applicability to military and wildlife scenarios. At the same time, the comparative analysis and participant feedback methods are used in [19] to evaluate strain-specific camouflage under operational conditions, while environmental testing and simulation is used in [19] to address challenges in military desert operations. The prototype validation and similarity analysis conducted in [26], image restoration and error correction deployed in [27], and the use of deception through decoys and imitation of biological traits [3] ensure effective target blending in military surveillance. Moreover, in [28], genetic algorithms optimise camouflage effectiveness through reaction time data, while techniques like Non-printability Score Loss (LNPS) and Total Variation Loss (LTV) in [30] and [31] are tailored for adversarial camouflage in urban military settings.

D. Generalisable methods

Statistical modelling, content-style loss curves, and colour quantisation techniques applied in [19] are examples of versatile approaches applicable across industries. CNN-based texture mapping and statistical data analysis deployed in [21] are used in diverse fields like image processing and generation. Furthermore, techniques like GLMM and GAN metrics discussed in [19] can be applied in user experience research and model training. Furthermore, prototyping and functional testing methods applied in [25] are standard in engineering and design. Similarly, target detection accuracy, cosine similarity, and error calculation used in [27] are common in various computer vision and general Machine Learning applications. At the same time, [29] employs colour mapping and CNN-based loss function optimisation, which are widely used in image processing and design tasks. Metrics such as MSE, UIQI, and confidence reduction applied in [30],

[31] have applications in adversarial AI and object detection. Additionally, [3] explores biological mimicry techniques that extend to various robotics, AI, and surveillance applications, demonstrating how performance metrics can be adapted for various fields beyond military applications.

V. EVALUATION METRICS

This section explores the evaluation metrics used for evaluating military camouflage designs. Building upon the categorisation of evaluation methods discussed in the previous section, this section focuses on the specific metrics used to assess the effectiveness of military camouflage design patterns. These metrics are categorised based on their nature (qualitative or quantitative) and their background (context-specific or generalisable).

A. Quantitative methods

The effectiveness of military camouflage designs can be quantitatively assessed through a variety of metrics. To this extent [18] include detection accuracy and spectral indicators, incorporating metrics on illumination, vegetation coverage, and colour coefficient to compare various terrain features. Furthermore, [18] assess the detectability of their solution by using a CNN to perform image recognition and assessing the loss and detection accuracy. Eye movement metrics such as the number of saccades, amplitude of the first saccade, and duration of the gaze offer insights into the effort required by an observer to extract information and distil a camouflaged target from a given background [20].

Additional metrics, such as detection time and hit rate, provide an inverse correlation between these metrics and camouflage effectiveness, with longer detection times (measured using the length an observer's gaze) and lower hit rates indicating superior camouflage [21]. Reaction time measurements further quantify the difficulty of detecting camouflaged objects [22], [23]. [24] developed a dual adversarial camouflage design which was validated against both human as well as AI observers. For human observers, a subjective experiment was conducted in which [24] measured the time spent and the overall accuracy of their inspections. At the same time, [24] uses metrics such as reaction times, GAN losses, perceptual and feature map losses, and saliency detection models to assess adversarial camouflage effectiveness and its impact on object detection systems like YOLOv5 and Faster-RCNN.

[31] evaluates adversarial attacks using confidence reduction metrics, model comparisons, and parameter optimisation. In a comparative approach, [26] focuses on similarity percentage analysis, performance testing of image merging algorithms, while [27] measures blending effectiveness with target detection accuracy and cosine similarity, and [28] builds a hybrid approach based on genetic algorithms and DNNs (deep neural networks) for reaction time-based optimisation and prediction of detection difficulty. In assessing the capabilities of their design to deceive AI observers the decrease in detection was measured. Advanced quantitative methods include loss function optimisation—utilising MSCN coefficients and feature maps— and error calculations based on a weighted combination cosine similarity [27], [29]. Behavioural metrics, like detection time and comparative analysis, measure speed and accuracy [24], [28]. Finally, metrics such as mean squared error (MSE) and universal image quality index (UIQI) provide objective measures of camouflage-background similarity [30].

B. Qualitative methods

Military camouflage designs can also be evaluated qualitatively through visual observation and subjective analysis. As such the criteria used in [19] include design complexity, reusability, practicality, adaptability, and material properties.

The paired comparison test is another qualitative metric, using visual assessments of digital images with camouflaged targets. Participants are asked to rate camouflage effectiveness on a Likert scale, providing comparative feedback across patterns [22]. Additionally, the ability to disrupt an object's silhouette — a key visual metric — enhances detectability reduction by making the target less recognisable [31].

C. Context-specific methods

Context-specific metrics provide nuanced evaluations tailored to specific environments or scenarios. For example, spectral indicators evaluate camouflage effectiveness by measuring spectral properties, such as winter wheat, on illumination. This includes assessing how well the applied systems blend into their surroundings using CNNs and spectral cameras like the FC200 on DJI Phantom drones [18].

Saliency detection models, such as the Reverse Attention Saliency detection model, simulate human vision to evaluate camouflage. By comparing saliency maps of original and camouflaged images, these models verify the effectiveness of designs in blending into their environment and deceiving human perception [24]. Metrics for environmental testing include energy efficiency, motor angles, and navigation performance of camouflaged robots, which evaluate functionality in specific operational settings [25].

Additionally, confidence reduction and ASR (Adversarial Success Rate) metrics are used to assess adversarial camouflage designs against machine learning models and object detection systems. These metrics measure how effectively adversarial designs lower a model's prediction confidence and quantify their real-world performance [31].

D. Generalisable methods

Colour quantisation techniques and CNN-based texture mapping are versatile methods used in image processing and design. These techniques are applicable in fields like

computer vision, where accurate colour representation and texture analysis are crucial [19], [21], [29].

Metrics such as detection time, hit rate, and eye movement analysis (e.g., saccades, gaze duration) provide insights into human perception and behaviour. These metrics are valuable in user studies, human-computer interaction research, and evaluating the effectiveness of visual designs [20], [21], [24]. Metrics such as cosine similarity, confidence reduction, and error calculations are commonly used in machine learning and computer vision. These metrics help in evaluating the accuracy and effectiveness of models and algorithms, making them applicable to a wide range of domains [24], [27], [30], [31].

VI. CONCLUSIONS

The evaluation of military camouflage design patterns involves various methods and metrics tailored to different contexts. By integrating qualitative approaches like visual observation with quantitative techniques such as detection accuracy, reaction time, and machine learning-based evaluations, researchers can capture both perceptual and operational dimensions of effectiveness.

However, there is a significant difference in how human observation and AI observation work in relation to the visual observation for instance. Human observers rely on perceptual cues and subjective interpretation, while AI systems use algorithmic processing and pattern recognition. This divergence means that current metrics often cater to one type of observation over the other, lacking a unified qualitative evaluation method that encompasses both perspectives.

Evaluation metrics are refined in relation to specific contexts and technologies used, including environmental conditions and adversarial challenges, while generalisable metrics provide versatile tools for various scenarios. The increasing use of AI enhances the evaluation of camouflage designs and the design process itself through background extraction techniques, such as real-time pattern generation. Adversarial patterns, designed to deceive both human and AI observers, are particularly relevant given the rise of autonomous systems, emphasizing the need for tailored evaluation methods.

Therefore, interdisciplinary collaboration is essential for developing effective dual-purpose camouflage evaluation methods and metrics that account for the full range of domain dynamics, including environmental, technological, and adversarial factors. By addressing the differences between human and AI observation, researchers can create more comprehensive and robust evaluation frameworks that improve the effectiveness of camouflage in diverse operational contexts.

REFERENCES

- [1] L. Talas, R. J. Baddeley, and I. C. Cuthill, "Cultural evolution of military camouflage," *Philosophical Transactions of the Royal Society B: Biological Sciences*, vol. 372, no. 1724, p. 20160351, 2017.
- [2] N. Katagiri, "Artificial Intelligence and Cross-Domain Warfare: Balance of Power and Unintended Escalation," *Global Society*, vol. 38, no. 1, pp. 34–48, 2024.
- [3] M. Ron and J. M. Thomas, "Military mimicry: the art of concealment, deception, and imitation," *Defense & Security Analysis*, vol. 40, no. 3, pp. 379–404, 2024.
- [4] A. Mondal, "Camouflage design, assessment and breaking techniques: a survey," *Multimedia Systems*, vol. 28, no. 1, pp. 141–160, 2022.

- [5] T. Alexander and H. Maarten, A., "Review of camouflage assessment techniques," *Target and background signatures*, no. 11536, p. 1153604, 2020.
- [6] L. Maozhen, "A military reconnaissance network for small-scale openscene camouflaged people detection," *Expert Systems*, vol. 40, no. 10, p. e13444, 2023.
- [7] M. Robert, F., "Camouflage Combat Uniform," *Defense AR Journal*, vol. 27, no. 4, pp. 354–397, 2020.
- [8] T. Laszlo, B. Ronald, and C. Innes, C., "Cultural evolution of military camouflage," *Philosophical Transactions of the Royal Society B: Biological Sciences*, vol. 372, no. 154, p. 20160351, 2020.
- [9] O. Boichak, "Camouflage aesthetics: militarisation, craftivism, and the in visibility of resistance at scale," *Contemporary Voices: St Andrews Journal of International Relations*, vol. 3, no. 1, p. 13, 2021.
- [10] P. Yogi, Tri, "Utilization of color similarity index for evaluating existing military camouflage designs," in *10th Annual International Conference on Industrial Engineering and Operations Management*, 2020, p. 9.
- [11] Y. Xin, X. Wei-dong, J. Qi, and L. Jun, "MF-CFI: A fused evaluation index for camouflage patterns based on human visual perception," *Defence Technology*, vol. 17, no. 5, pp. 1602–1608, 2021.
- [12] H. Bentian, X. Weidong, Y. Xin, X. Feifei, L. Hao, and H. Wei, "Research on military targets camouflage effect rating based on background anomaly detection and deep transfer features learning," *Journal of Industrial Textiles*, vol. 54, pp. 1–24, 2024.
- [13] H. Kyo-Seong and M. Jungmok, "Military camouflaged object detection with deep learning using dataset development and combination," *The Journal of Defense Modeling and Simulation*, pp. 1–12, 2024.
- [14] S. Ali, T. Abuhmed, S. El-Sappagh, K. Muhammad, J. M. Alonso-Moral, R. Confalonieri, R. Guidotti, J. Del Ser, N. D'iaz-Rodríguez, and F. Herrera, "Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence," *Information fusion*, vol. 99, p. 101805, 2023.
- [15] R. Den Hollander, A. Adhikari, I. Tolios, M. van Bekkum, A. Bal, S. Hendriks, M. Kruihof, D. Gross, N. Jansen, G. Perez *et al.*, "Adversarial patch camouflage against aerial detection," *SPIE Security*, vol. 11543, pp. 77–86, 2020.
- [16] P. Peter, G. Simon, O. Aidan, C. Michael, and H. Dan, "AI deception: A survey of examples, risks, and potential solutions," *Patterns*, vol. 5, no. 5, 2020.
- [17] D. Moher, A. Liberati, J. Tetzlaff, and D. G. Altman, "Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement," *Journal of Clinical Epidemiology*, vol. 62, no. 10, pp. 1006–1012, 2009.
- [18] N. A. Pasichnyk, S. A. Shvorov, O. A. Opryshko, A. Dudnyk, B. Oksana, and D. S. Komarchuk, "Prospective of Camouflage Ground Objects to Counter UAVs," in *2023 IEEE 7th International Conference on Methods and Systems of Navigation and Motion Control (MSNMC)*. Ieee, 2023, pp. 1–6.
- [19] Q. Jia, C. FMa, L. Zhu, and H. Xue, "A Comparative Study on the Realization Methods of Computer Aided Design Facial Camouflage," in *2020 IEEE International Conference on Advances in Electrical Engineering and Computer Applications (AEECA)*. Ieee, 2020, pp. 753–757.
- [20] X. Wei, G. Li, and K. Wang, "A novel method for automatic camouflage pattern synthesize," *IEEE Access*, vol. 9, pp. 67 559–67 568, 2021.
- [21] X. Wei, K. Wang, G. Li, and H. Kim, "An automatic design of camouflage patterns based on CNNs," in *Proceedings of the 2020 6th International Conference on Computing and Artificial Intelligence*, 2020, pp. 257–260.
- [22] E. Daneshvar, M. A. Tehran, and Y.-J. Zhang, "Hybrid camouflage pattern generation using neural style transfer method," *Color Research & Application*, vol. 47, no. 4, pp. 878–891, 2022.
- [23] L. Talas, J. Fennell, K. Kjernsmo, I. Cuthill, N. Scott-Samuel, and R. Baddeley, "CamoGAN: evolving optimum camouflage with Generative Adversarial Networks," *Methods in Ecology and Evolution*, vol. 11, no. 2, pp. 240–247, 2020.
- [24] Y. Wang, T. Cao, Y. Zheng, Y. Wang, L. Chen, B. Fu, and T. Han, "A generation method of dual adversarial camouflage," in *Proceedings of the 2022 6th International Conference on Electronic Information Technology and Computer Engineering*, 2022, pp. 1069–1074.
- [25] M. Zouai, A. Merizig, I. Boudjelkha, H. Belouaar, O. Kazar, G. Ortiz, A. Lakas, and Z. Houhamdi, "Design and Development of an IoT-based Solar Powered Camouflaged Robot for Military Applications," in *2022 International Arab Conference on Information Technology (ACIT)*. Ieee, 2022, pp. 1–6.
- [26] R. A. Reddy, S. W. Ahmed, P. S. Santosh, J. K. Raj, and R. Balaji, "Image Merging Technique for a Camouflage Surveillance Robot," in *2023 IEEE 5th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA)*. Ieee, 2023, pp. 01–07.
- [27] Y. Xue, D. Zhang, Y. Zhang, and L. Li, "Research on Adaptive Camouflage System Based on Deep Learning," in *2023 3rd International Conference on Electronic Information Engineering and Computer (EIECT)*. Ieee, 2023, pp. 400–403.
- [28] J. G. Fennell, L. Talas, R. J. Baddeley, I. C. Cuthill, and N. E. Scott-Samuel, "The Camouflage Machine: Optimizing protective coloration using deep learning with genetic algorithms," *Evolution*, vol. 75, no. 3, pp. 614–624, 2021.
- [29] X. Cai, H. Li, and S. Guo, "Intelligent camouflage pattern generating in internet of things," *Internet Technology Letters*, vol. 6, no. 5, p. e349, 2023.
- [30] J. Kim, K. Lee, H. Lee, H. Yang, and S.-Y. Oh, "Camouflaged adversarial attack on object detector," in *2021 21st International Conference on Control, Automation and Systems (ICCAS)*. Ieee, 2021, pp. 613–617.
- [31] H. Farlow, M. Garratt, G. Mount, and T. Lynar, "The race to robustness: exploiting fragile models for urban camouflage and the imperative for machine learning security," in *2023 IEEE Region 10 Symposium (TENSYP)*. Ieee, 2023, pp. 1–6.